

Computational Physics Classes No.1 - Report

Simulation of projectile trajectory with Forward Euler method

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October 14, 2025

Abstract

This report presents a numerical study of projectile motion using the Forward Euler integration method. The simulation includes the effects of air resistance and its dependence on altitude. Results show that the Euler scheme achieves first-order accuracy and reproduces the expected physical behavior: air drag reduces both the range and height of the projectile, while decreasing air density with altitude extends its flight distance. The study demonstrates the effectiveness of simple numerical methods in modeling realistic projectile dynamics.

1 Introduction

The motion of a projectile is one of the fundamental problems in classical mechanics and serves as a natural example for testing and verifying numerical integration schemes. While the analytical solution of a projectile's trajectory is straightforward in the absence of air resistance, the inclusion of drag forces makes the equations of motion nonlinear and often unsolvable by analytical means. In such cases, numerical methods become indispensable tools for obtaining approximate solutions.

The goal of this project was to simulate the two-dimensional trajectory of a projectile launched with an initial velocity v_0 at an angle Θ_0 above the horizontal axis. The simulation was performed using the **Forward Euler method**. The study consists of several stages, starting with a trajectory in vacuum (no drag), then progressively including the effects of air resistance and its dependence on altitude.

2 Description of the Numerical Method

The trajectory of a projectile subject to gravity and drag can be described by Newton's second law in two dimensions:

$$m \frac{dv_x}{dt} = F_x, \quad (1)$$

$$m \frac{dv_y}{dt} = F_y, \quad (2)$$

where m is the mass of the projectile, and F_x, F_y are the components of the total force acting on it. In this project, the forces are defined as:

$$F_x = -Dvv_x \left(1 - \frac{ay}{T_0}\right)^\alpha, \quad (3)$$

$$F_y = -mg - Dvv_y \left(1 - \frac{ay}{T_0}\right)^\alpha, \quad (4)$$

where $v = \sqrt{v_x^2 + v_y^2}$ is the magnitude of the velocity vector, D is a drag coefficient, a is the coefficient describing the temperature dependence with altitude, T_0 is the reference temperature at ground level, g is the gravitational acceleration, and α is a constant describing how air density changes with altitude. When $D = 0$, the equations reduce to the well-known vacuum case with purely parabolic motion.

The numerical integration was performed using the **Forward Euler method**. For each time step Δt , the updated positions and velocities are computed as:

$$x_{n+1} = x_n + \Delta t v_{x,n}, \quad (5)$$

$$y_{n+1} = y_n + \Delta t v_{y,n}, \quad (6)$$

$$v_{x,n+1} = v_{x,n} + \frac{\Delta t}{m} F_{x,n}, \quad (7)$$

$$v_{y,n+1} = v_{y,n} + \frac{\Delta t}{m} F_{y,n}. \quad (8)$$

This algorithm approximates the continuous change of position and velocity by assuming that the acceleration remains constant during each time interval Δt . Although the Forward Euler method is only first-order accurate (the local truncation error scales as $\mathcal{O}(\Delta t^2)$, and the global error as $\mathcal{O}(\Delta t)$), it provides a simple and clear framework for investigating the qualitative behavior of projectile motion and for assessing numerical convergence.

The numerical simulation proceeds until the projectile returns to the ground level ($y < 0$). At that moment, a linear interpolation between the last two time steps is performed to correct the impact position and obtain a more accurate estimate of the range $x_{\max}$. This correction helps reduce the discretization error at the collision point and ensures better agreement with the analytical solution in the vacuum case.

3 Results

3.1 Projectile Motion in Vacuum (No Drag)

The first step of the study consisted of simulating the projectile motion in the absence of air resistance, i.e. with the drag coefficient $D = 0$. Under these conditions, the trajectory of the projectile should follow an ideal parabolic path, for which the analytical expressions for the range and the time of flight are given by:

$$R_{\text{analytic}} = \frac{v_0^2 \sin(2\Theta_0)}{g}, \quad (9)$$

$$t_{\text{flight}} = \frac{2v_0 \sin(\Theta_0)}{g}. \quad (10)$$

These analytical results provide a natural benchmark for testing the accuracy of the numerical integration scheme. To investigate the influence of the integration time step Δt on the accuracy of the Forward Euler method, the simulation was repeated for several values of $n = \{10, 20, 50, 100, 200, 500\}$, where n denotes the number of integration steps over the total flight time.

Figure 1 shows the calculated trajectories for these different values of n . As expected, for coarse time steps (e.g. $n = 10$ or $n = 20$), the numerical solution deviates visibly from the analytical curve, especially near the apex of the trajectory and close to the impact point. As the number of time steps increases, the results converge smoothly toward the analytical parabola, demonstrating the first-order accuracy of the Forward Euler algorithm.

The simulation also reveals that the Euler method slightly underestimates the projectile's maximum height and range for large time steps. This behavior is characteristic of explicit integration schemes, where truncation errors accumulate over time and cause a systematic loss of energy in the numerical trajectory. Nevertheless, for $n \geq 200$, the difference between the analytical and numerical results becomes negligible, indicating sufficient accuracy for further simulations involving drag forces.

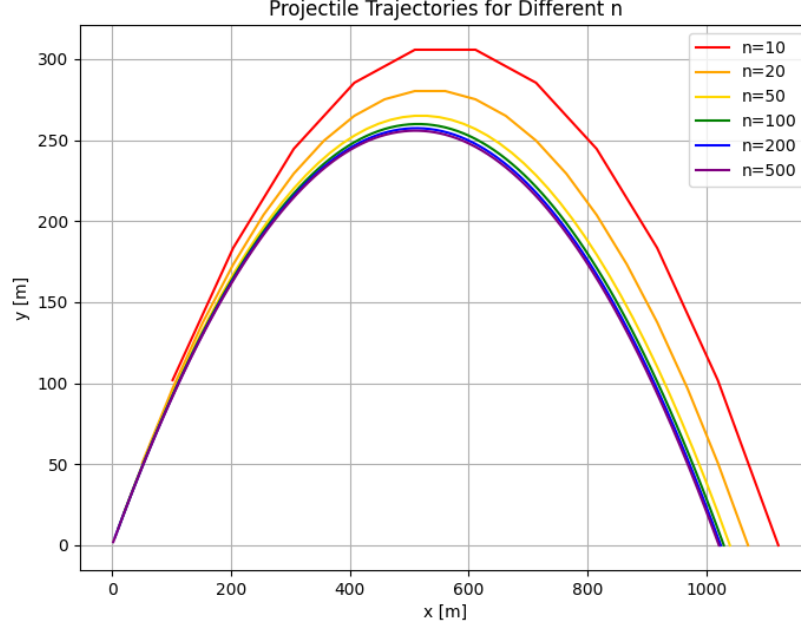


Figure 1: Projectile trajectories simulated using the Forward Euler method for different numbers of integration steps n in the absence of drag force.

3.2 Accuracy and Global Error Analysis

In order to quantify the accuracy of the Forward Euler method, the *global error* at the moment of impact was analyzed. The global error E_{glob} is defined as the absolute difference between the analytical range R_{analytic} and the numerically computed range $x_{\text{num,max}}$:

$$E_{\text{glob}} = |R_{\text{analytic}} - x_{\text{num,max}}|. \quad (11)$$

For each simulation, the time step was chosen as $\Delta t = t_{\text{flight}}/n$, where n is the number of integration intervals. The resulting errors were then plotted as a function of Δt on a logarithmic scale (see Fig. 2). The plot clearly shows that the global error decreases as the time step becomes smaller, which is consistent with the theoretical behavior of the Forward Euler scheme.

In particular, for sufficiently small Δt , the error exhibits an approximately linear dependence on the time step in the log-log scale, i.e.

$$E_{\text{glob}} \propto \Delta t,$$

confirming that the method is **first-order accurate**. This means that halving the time step roughly halves the global error. At larger Δt values, deviations from the linear trend appear due to the accumulation of truncation errors and the limited resolution of the trajectory near the ground impact, where the interpolation correction becomes less accurate.

Overall, the numerical results agree well with theoretical expectations: the Forward Euler method provides a simple yet reliable approach for solving projectile motion, but achieving high precision requires sufficiently small time steps because of its relatively low order of convergence.

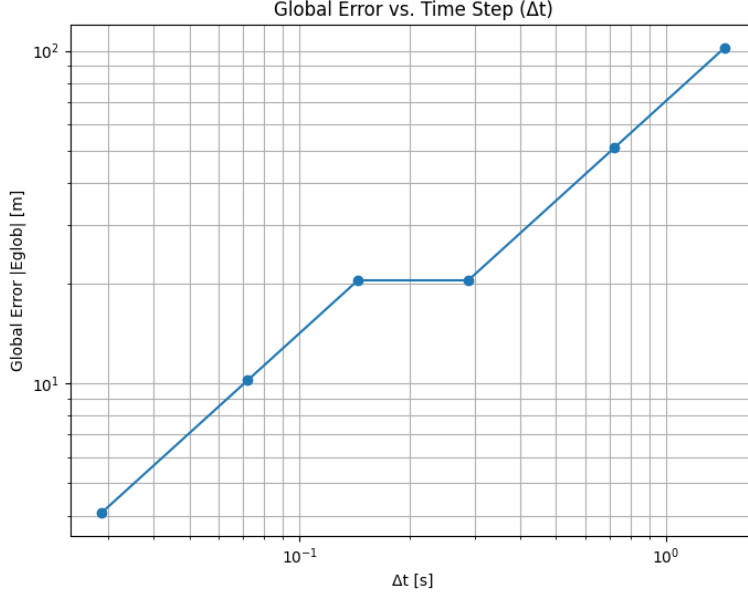


Figure 2: Global error of the projectile’s range as a function of time step Δt for the Forward Euler method in vacuum. The nearly linear relationship on the log–log scale confirms the first–order convergence of the method.

3.3 Projectile Motion with Constant Air Density

In the next stage of the simulation, the air resistance was taken into account while the dependence of air density on altitude was temporarily neglected by setting the parameter $a = 0$. Under these conditions, the drag force acts with constant magnitude relative to the velocity squared, and its components are expressed as:

$$F_x = -Dvv_x, \quad (12)$$

$$F_y = -mg - Dvv_y, \quad (13)$$

where D is the drag coefficient and $v = \sqrt{v_x^2 + v_y^2}$ is the instantaneous velocity magnitude. The drag term introduces a nonlinear damping proportional to the square of the velocity, which reduces both the maximum height and the horizontal range of the projectile.

The trajectories were computed for several drag coefficients $D = \{0, 1 \times 10^{-4}, 2 \times 10^{-4}, 5 \times 10^{-4}, 1 \times 10^{-3}\}$, keeping all other parameters constant. The results are presented in Fig. 3. As expected, the inclusion of drag significantly alters the projectile motion. With increasing D , the trajectory becomes shorter and more curved, indicating a faster reduction of horizontal velocity due to air resistance. The projectile reaches a lower maximum altitude and a smaller horizontal range, while its descent becomes steeper.

It can be observed that even relatively small drag coefficients cause a noticeable deviation from the ideal parabolic path. For $D = 1 \times 10^{-3}$, the range is reduced by nearly half compared to the vacuum case, demonstrating the strong influence of aerodynamic drag on projectile motion. The dependence of the range on D is nonlinear, as both the vertical and horizontal components of velocity are continuously reduced during the flight.

Overall, these results confirm that the implementation of the drag force in the numerical model behaves as expected and reproduces the physically realistic damping of motion due to air resistance.

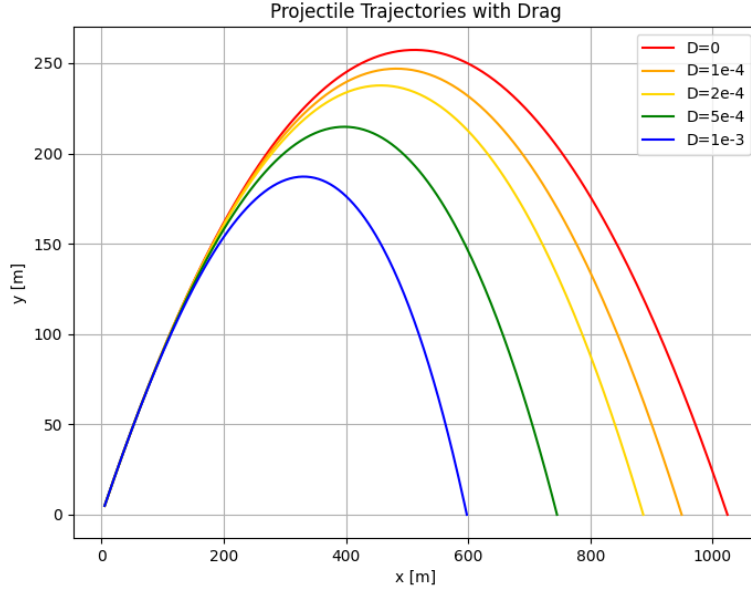


Figure 3: Projectile trajectories calculated for different drag coefficients D with altitude dependence turned off ($a = 0$). Increasing drag reduces both the range and the maximum height of the projectile.

3.4 Dependence of Range on Launch Angle and Drag

In this part of the project, the dependence of the projectile range on the launch angle was investigated for different values of the drag coefficient D . The simulation was performed for firing angles in the range $\Theta_0 \in [15^\circ, 65^\circ]$ with a fixed initial velocity $v_0 = 100$ m/s. Three representative drag coefficients were analyzed: $D = 0$ (no drag), $D = 1 \times 10^{-3}$, and $D = 2 \times 10^{-3}$. For all cases, the altitude correction was turned off ($a = 0$) to isolate the pure influence of air resistance on the trajectory.

Figure 4 shows the range as a function of the firing angle for $D = 0$, i.e. for ideal projectile motion without drag. The curve is symmetric and reaches its maximum at $\Theta_0 = 45^\circ$, which agrees perfectly with the analytical prediction:

$$R_{\max} = \frac{v_0^2}{g}.$$

This confirms the correctness of the numerical procedure and provides a baseline for comparison with cases that include aerodynamic drag.

When drag is introduced (Figs. 5 and 6), the overall range decreases significantly and the symmetry of the curve is lost. The optimal firing angle, corresponding to the maximum range, shifts to smaller values—approximately 40° for $D = 1 \times 10^{-3}$ and around 37° for $D = 2 \times 10^{-3}$. This shift arises because the projectile experiences a larger horizontal deceleration at higher angles, which increases the time it spends in flight and hence the total energy lost to air resistance.

The observed behavior is physically consistent with real projectile motion in air: drag shortens the trajectory and makes the optimal angle for maximum range smaller than 45° . Additionally, the reduction in range becomes increasingly pronounced for larger D , as the resistive force scales quadratically with velocity.

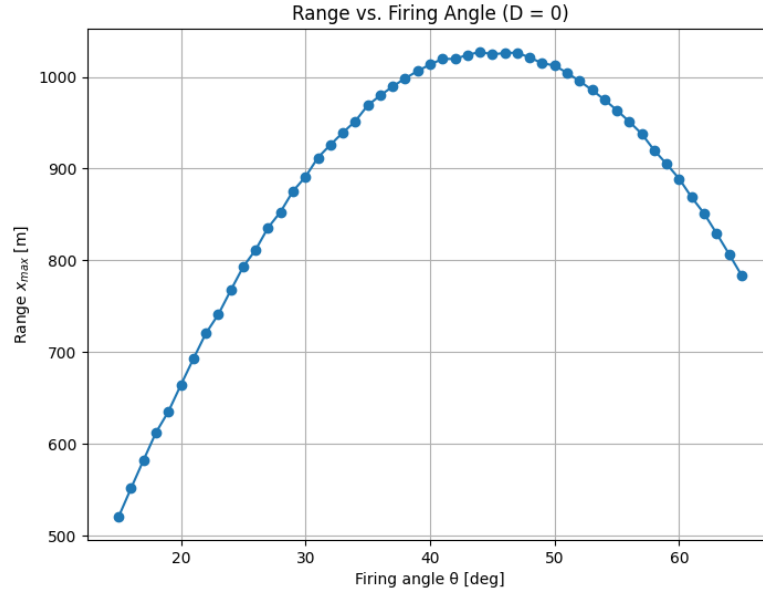


Figure 4: Range of the projectile as a function of the launch angle in the absence of drag ($D = 0$). The maximum occurs at $\Theta_0 = 45^\circ$, consistent with the analytical solution.

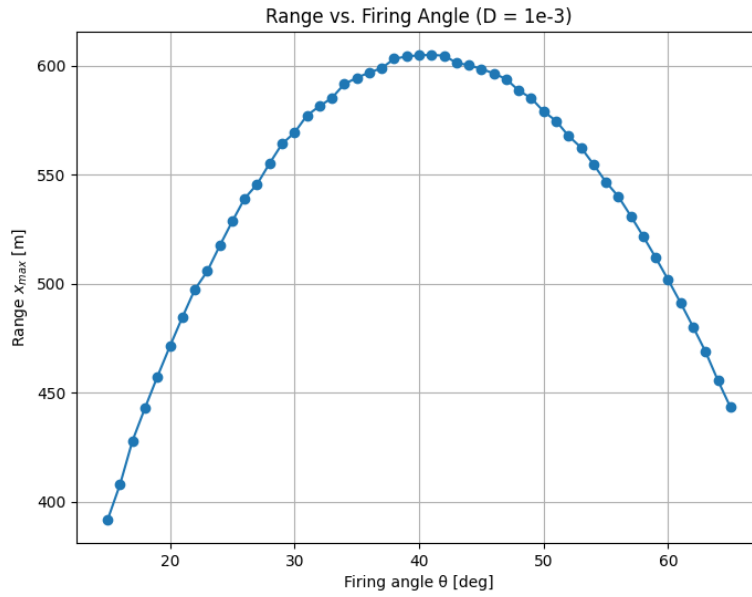


Figure 5: Range as a function of firing angle for $D = 1 \times 10^{-3}$. The presence of drag reduces the range and shifts the optimal angle to about 40° .

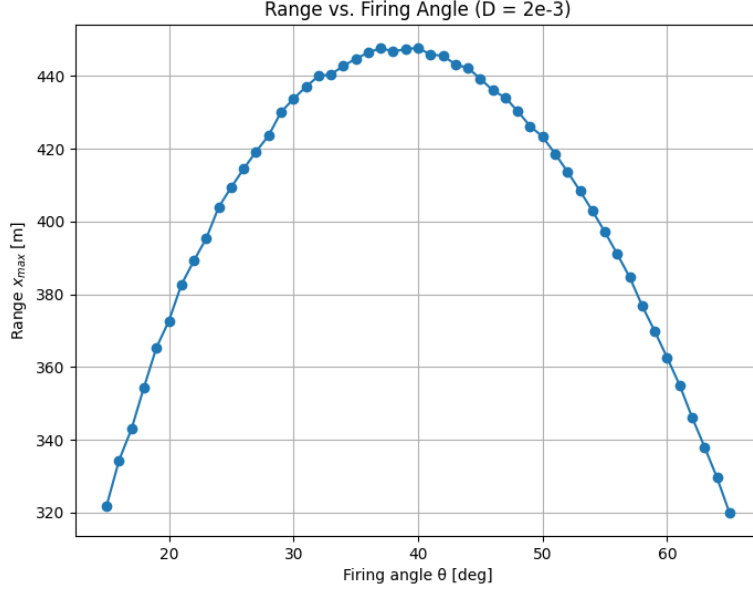


Figure 6: Range as a function of firing angle for $D = 2 \times 10^{-3}$. Stronger drag further shortens the range and lowers the optimal angle to approximately 37° .

3.5 Projectile Motion with Altitude-Dependent Air Density

In the final stage of the project, the drag force was extended to include the effect of decreasing air density with altitude. This correction is introduced through the factor

$$\left(1 - \frac{ay}{T_0}\right)^\alpha,$$

which modifies the drag term and effectively reduces the magnitude of aerodynamic resistance as the projectile rises. Here, $a = 6.5 \times 10^{-3}$ represents the temperature lapse rate, $T_0 = 293$ K is the reference ground temperature, and $\alpha = 2.5$ controls the rate of density variation.

The simulations were performed for a high-velocity projectile ($v_0 = 700$ m/s, $m = 20$ kg) with a constant drag coefficient $D = 10^{-3}$. Two firing angles (35° and 45°) were examined, each with and without the altitude correction ($a = 0$ and $a = 6.5 \times 10^{-3}$). The resulting trajectories are shown in Fig. 7.

As expected, introducing the altitude correction increases both the range and the maximum height of the projectile. Because air density decreases with altitude, the drag force weakens as the projectile ascends, allowing it to maintain a higher velocity for a longer time. The effect is especially noticeable for large initial speeds, where the projectile reaches several kilometers of altitude and thus experiences significantly thinner air.

The trajectories corresponding to $a = 0$ (red and orange curves) display a stronger damping and shorter range compared to the corrected cases (yellow and green curves). The difference between the two firing angles is also amplified: at 45° , the altitude effect contributes more substantially to the increased range due to the longer flight time and higher maximum altitude.

Overall, this final test demonstrates that the implemented model correctly captures the physical influence of atmospheric density variation on projectile motion. The numerical results are consistent with real ballistic behavior, where long-range projectiles benefit from reduced air resistance at higher altitudes.

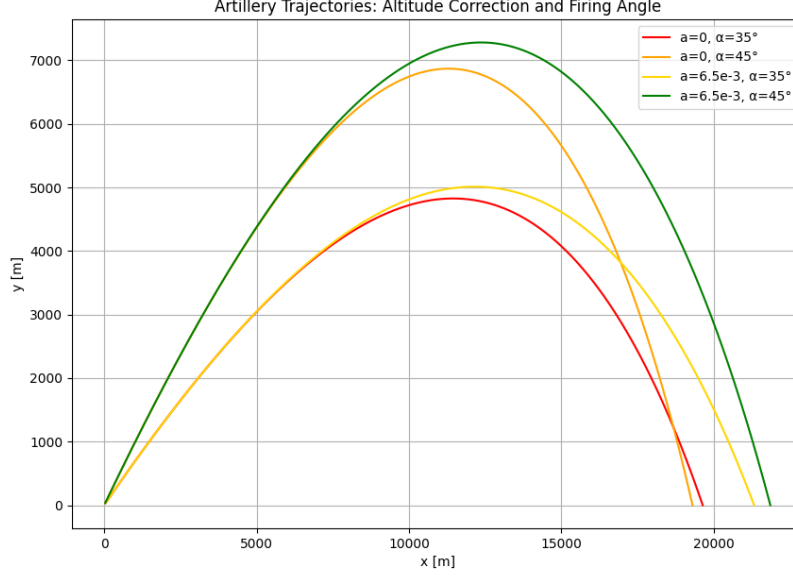


Figure 7: Artillery trajectories simulated for different firing angles and altitude-dependent drag corrections. The reduction of air density with altitude ($a = 6.5 \times 10^{-3}$) leads to longer ranges and higher apogees compared to the constant-density case ($a = 0$).

4 Conclusions

In this project, the Forward Euler method was successfully applied to simulate the two-dimensional motion of a projectile under various physical conditions. The comparison between numerical and analytical results confirmed that the method exhibits first-order accuracy: the global error decreases linearly with the integration time step Δt . For sufficiently small time steps, the numerical trajectories closely matched the analytical parabolas.

Introducing air resistance significantly altered the projectile's motion, reducing both its range and maximum height. The dependence of the range on the drag coefficient D was nonlinear, and the optimal firing angle shifted from 45° (in vacuum) to smaller values as drag increased. This behavior is consistent with physical expectations for motion through a resistive medium.

When the altitude correction was enabled, accounting for the decrease of air density with height, the projectile achieved greater ranges and higher apogees, particularly at large initial velocities. The implemented model thus captures the key qualitative and quantitative effects of aerodynamic drag and atmospheric variation.

Overall, the results demonstrate the effectiveness of simple numerical integration methods such as Forward Euler for studying projectile dynamics, while also highlighting their limitations in accuracy and stability for larger time steps.

References

- [1] Dr. Hab. Eng. Chwiej, T. *Project: Simulation of projectile trajectory with Forward Euler method..* AGH University of Krakow. Retrieved from https://galaxy.agh.edu.pl/~chwiej/comp_phys/labs/1_projectile_launch.pdf